

Flash Card 1 — Introduction

What to say

Hello everyone.

Today our team will present our project:

“Network Reliability under Edge Failures using Monte Carlo Simulation.”

In this project we study how **network structure affects reliability when connections randomly fail**, using both **graph theory** and **computer simulations**.

Flash Card 2 — Motivation

Many real-world systems can be modelled as networks.

Examples include:

- communication networks
- transportation systems
- power grids
- computer networks

In these systems, **links can fail randomly**.

(The basic example) Our question is:

How likely is the network to remain connected when edges fail?

This is called **network reliability**.

Flash Card 3 — Key Idea

What to say

Network reliability depends heavily on the **structure of the graph**.

We will explore:

- **Bridges (cut-edges)**

These structures determine **how easily a network can become disconnected**.

Our project studies the relationship between these structures and reliability.

Flash Card 4 — Graph Theory Background

What to say

First we review some basic graph theory.

A **cut-edge**, also called a **bridge**, is an edge whose removal disconnects the graph.

This means if that edge fails, the network immediately breaks into two parts.

A **minimum cut** is the smallest set of edges whose removal disconnects the graph.

The size of this set is called λ (**lambda**).

The larger λ is, the more reliable the network tends to be.

What to say

Our theory section focuses on three key results.

First:

If a **cut-edge fails**, the graph becomes disconnected.

Second:

If a graph has **k bridges**, the reliability is at most

$(1-q)^k$ where q is the edge failure probability.

Third:

If the **minimum cut size is λ** , the disconnection probability is proportional to

q^λ

These results connect **graph structure with reliability behaviour**.

Flash Card 6 — What is Monte Carlo simulation and why ?

What to say

For large networks, calculating reliability exactly becomes very difficult.

So we use a computational approach called **Monte Carlo simulation**.

What is Monte Carlo Simulation is? And explain by example : Pi

The key idea in network reliability using **m.c.s** in graphs is simple:

1. Randomly fail edges with probability P
2. Check if the graph remains connected
3. Repeat this many times
4. Estimate reliability from the results

This allows us to approximate reliability experimentally.

Flash Card 7 — Algorithm

What to say

Our simulation algorithm works as follows:

1. Generate a graph and We tested four types of graphs.

First, **tree graphs**.

A tree has many bridges, so **any edge failure disconnects the graph**.
This makes it very fragile.

Second, **cycle graphs**.

A cycle requires **two failures** to disconnect, so it is more robust.

Third, **complete graphs**.

These graphs have very high connectivity and are the **most robust networks, at least edges: $n-1$**

Fourth, **degree constrained graph: $3 \leq \deg(v) < n-2$**

More reality: graphs **in between cycle and complete** graphs

For each graphs:

2. For each edge, randomly decide whether it fails
3. Test if the resulting graph is still connected

4. Repeat thousands of times
5. Compute the proportion of connected outcomes

This gives us an estimate of **network reliability**.

Flash Card 8 — Experimental Graphs

What to say

Our simulations produced several useful plots.

We analyzed:

- Reliability vs failure probability
- Estimated Reliability: number of connected trials / total number of trials

The results confirmed our theoretical predictions.

Graphs with **more bridges are less reliable**. This confirms that **graph structure strongly determines network reliability**.

Monte Carlo simulation provides a practical way to verify these results.

Flash Card 10 — Conclusion

What to say

To summarize:

- Network reliability depends on **graph structure**
complete: strongest (stablest), degree constrained(stable, reality)
- Monte Carlo simulation provides an effective method to estimate reliability

Our project shows how **mathematics and computation work together** to analyze complex networks.

Flash Card 11 — Closing

What to say

Thank you for listening.

We would be happy to answer any questions.